

Generation Interconnection System Impact Study Report

For Queue Project Q14-0313

Clairemont 345 kV

June 01, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0313. The Interconnection Customer proposes a 60.2 MW generating facility interconnecting at the Clairemont 345 kV substation in the Kentucky area. The request was received on March 01, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 201 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$43,879,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Alibates to Latah Jct 138.0 kV
Contingency: Frontier to Harvest Power Orlando Substation 138.0 kV
Loading: 116.9% (Rating: 322 MVA)
DFAX: 53.7%
Pre-project loading: 106.9%
2. Facility: Harvest Power Orlando to Frontier 138.0 kV
Contingency: Spatsum Lornex to Latah Jct 138.0 kV line
Loading: 116.2% (Rating: 304 MVA)
DFAX: 50.6%
Pre-project loading: 106.2%
3. Facility: Hare Substation to N. Shannon 138.0 kV line
Contingency: Weber to Lark Ellen 138.0 kV
Loading: 114.0% (Rating: 214 MVA)
DFAX: 11.1%
Pre-project loading: 110.9%
4. Facility: Frontier to Yuba City Sta A 138.0 kV
Contingency: McClellan Ar to Spatsum Lornex 138.0 kV
Loading: 113.7% (Rating: 405 MVA)
DFAX: 77.1%
Pre-project loading: 102.3%
5. Facility: Westin-Elgin to Yuba City Sta A 138.0 kV
Contingency: Spatsum Lornex Substation to McClellan Ar 138.0 kV
Loading: 112.7% (Rating: 418 MVA)
DFAX: 77.1%
Pre-project loading: 101.6%
6. Facility: Spatsum Lornex to Latah Junction 138.0 kV
Contingency: Harvest Power Orlando to Frontier 138.0 kV
Loading: 110.8% (Rating: 349 MVA)

DFAX: 53.7%

Pre-project loading: 101.5%

7. Facility: McClellan Ar to Westin-Elgin 138.0 kV line
Contingency: Yuba City Sta A to Westin-Elgin Substation 138.0 kV
Loading: 109.5% (Rating: 359 MVA)
DFAX: 59.7%
Pre-project loading: 99.5%
8. Facility: Country Club 2 to Kettle 138.0 kV
Contingency: Spatsum Lornex to Latah Jct 138.0 kV
Loading: 107.5% (Rating: 154 MVA)
DFAX: 32.5%
Pre-project loading: 94.8%
9. Facility: Lark Ellen to Laguna Tap 138.0 kV
Contingency: N. Shannon to Hare 138.0 kV Line
Loading: 107.2% (Rating: 215 MVA)
DFAX: 10.0%
Pre-project loading: 104.4%
10. Facility: McClellan Ar to Spatsum Lornex 138.0 kV line
Contingency: Yuba City Sta A to Westin-Elgin 138.0 kV
Loading: 107.1% (Rating: 464 MVA)
DFAX: 81.4%
Pre-project loading: 96.5%
11. Facility: Lark Ellen to Weber 138.0 kV
Contingency: Hare to N. Shannon 138.0 kV
Loading: 106.9% (Rating: 220 MVA)
DFAX: 10.0%
Pre-project loading: 104.1%
12. Facility: Temple Sta 243 Substation to Westin-Elgin 138.0 kV
Contingency: McClellan Ar to Westin-Elgin 138.0 kV
Loading: 106.7% (Rating: 300 MVA)
DFAX: 44.4%
Pre-project loading: 97.8%
13. Facility: Studio to Laguna Tap Substation 138.0 kV Line
Contingency: Hare to N. Shannon 138.0 kV line
Loading: 105.9% (Rating: 213 MVA)
DFAX: 10.0%
Pre-project loading: 103.1%
14. Facility: North Little Rock Levy to E Sanger 138.0 kV
Contingency: N. Shannon to Hare 138.0 kV
Loading: 103.1% (Rating: 216 MVA)
DFAX: 5.6%
Pre-project loading: 101.5%
15. Facility: Portage Lakes to Old Forge 230.0 kV
Contingency: Surrowiec to Weedpatch 500.0 kV
Loading: 102.0% (Rating: 694 MVA)
DFAX: 5.4%
Pre-project loading: 101.5%
16. Facility: E Sanger to Truman Tap Substation 138.0 kV
Contingency: N. Shannon to Hare 138.0 kV
Loading: 100.4% (Rating: 210 MVA)
DFAX: 5.6%

Pre-project loading: 98.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alibates to Latah Jct 138.0 kV	Rebuild line with higher-capacity conductor	\$14,939,000	38% (\$5,674,000)
Frontier to Harvest Power Orlando 138.0 kV	Rebuild line with higher-capacity conductor	\$13,424,000	48% (\$6,431,000)
Hare Substation to N. Shannon 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,194,000	4% (\$174,000)
Frontier to Yuba City Sta A Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$12,977,000	61% (\$7,928,000)
Westin-Elgin to Yuba City Sta A 138.0 kV	Rebuild line with higher-capacity conductor	\$17,870,000	61% (\$10,917,000)
Spatsum Lornex to Latah Junction Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$7,639,000	38% (\$2,901,000)
Mcclellan Ar to Westin-Elgin 138.0 kV line	Reconductor line	\$2,358,000	60% (\$1,420,000)
Kettle to Country Club 2 138.0 kV	Reconductor line	\$3,832,000	100% (\$3,832,000)
Lark Ellen to Laguna Tap 138.0 kV	Reconductor line	\$966,000	4% (\$41,000)
Mcclellan Ar to Spatsum Lornex 138.0 kV line	Reconductor line	\$4,851,000	60% (\$2,921,000)
Lark Ellen to Weber 138.0 kV	Reconductor line	\$4,941,000	4% (\$211,000)
Westin-Elgin to Temple Sta 243 138.0 kV	Reconductor line	\$788,000	57% (\$451,000)
Studio to Laguna Tp 138.0 kV	Reconductor line	\$3,020,000	4% (\$129,000)
N Little Rock Levy to E Sanger 138.0 kV	Reconductor line	\$2,308,000	6% (\$132,000)
Portage Lakes to Old Forge 230.0 kV	Reconductor line	\$6,843,000	7% (\$452,000)
E Sanger to Truman Tap Substation 138.0 kV	Reconductor line	\$4,609,000	6% (\$265,000)
	Total allocated to project		\$43,879,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.